MIL-F-48425A(AR)
AMENDMENT 2
9 May 1983
SUPERSEDING
AMENDMENT 1
8 May 1982

MILITARY SPECIFICATION

FIN ASSEMBLY, M24

This Amendment forms a part of Military Specification MIL-F-48425A(AR), dated 11 September 1980, and is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.2, Delete: "ASTM E-18, Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.

PAGE 3

3.4, Delete: "fin housing" where it appears (2 places), and substitute "cartridge housing".

PAGE 4

4.3.1: Add the following to the list of components submitted for first article testing:

	Part Description	Drawings	Quantity
	Flat tensile specimen	See 4.5.3.2	2
	Round tensile specimen	See 4.5.3.2	2
	Flat fully prepared microstructure specimen	See 4.5.3.3.1	2
	Round fully prepared microstructure specimen	See 4.5.3.3.2	2
	Macroetch specimens from 1 extruded bar of housing stock	See 4.5.2	2
	Macroetch specimens from 1 extruded bar of fin stock	See 4.5.2	2
*	Add: "Blade security specimens	See 4.5.6	5"

Last 2 paragraphs: Delete in their entirety and substitute, "Macroetch specimens shall be identified as to their origin (front or rear of billet)."

FSC 1310

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PAGE 5

Table I, First Article Inspection:

Delete "Hydrotest, to destruction, 10 (a), 3.2, 4.5.4"

Delete "Extrusion Blanks (housing)", and substitute, "Fully heat-treated cartridge housing stock"

Delete "Hardness test 2(c), 3.1, 4.5.3.1"

Microstructure examination: Delete "1, 3.3, 4.5.3.3", and substitute "2(C), 3.3, 4.5.3.3.2"

Delete "Extrusion Blanks (Fin)" and substitute "Fully heat-treated fin stock"

Add: "Microstructure examination, 2, 3.3, 4.5.3.3.1"

Note a: Delete in its entirety

Note c: Delete in its entirety and substitute, "(C) Microstructure specimens shall be taken from either end of each round tensile specimen submitted for testing"

Note d: Delete in its entirety and substitute "(d) Largest standard flat-type specimens available"

PAGE 6

4.4.1.1:

Line 7, delete "more than one" and substitute "not more than three".

Delete "hardness" wherever it appears (2 places)

Delete last sentence from "however ..."

4.4.1.2 Fin stock lot. Delete in its entirety and substitute:

"4.4.1.2 Fin stock lot. The fin stock lot shall consist of not more than 2,000 pounds of extruded stock of the same alloy that has been thermally treated by the same method in the same furnace equipment at the same temperature and time in one unchanged process regardless of the solution heat treating method utilized. Each fin stock lot representing

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material that has been extruded and solution heat treated must be solution heat treated at the same time in the same furnace using the same temperature and time in that furnace. The fin stock lot thus constituted shall be the basis for selecting the sample blanks for the tension and microstructure tests hereinafter prescribed.

Fin stock lots produced by the press-quench method may be up to 2,000 pounds each or fraction thereof. The tensile test and microstructure specimens shall be taken from the beginning and end of the fin stock lot representing the first and last extruded material in the lot (except cropped areas).

PAGE 7

4.4.2.b: Add the following: "For inspection of para. 4.4.2.4 Minor 205 (Length of rear thread, max.): Inspect 1 housing at the end of each day's production by cutting the rear of the housing in half (longitudinally). Length shall be measured from the rear face to the end of the thread's run out. Any housing which does not meet the drawing requirement shall cause the rejection of all material produced since the last successful inspection."

PAGE 8

4.4.2.1: Delete "Fin and Housing Extrusion Bars" and substitute "Fin and Housing Extrusion bars prior to the final heat treatment."

PAGE 9

4.4.2.2: Delete "Fin and Housing Extrusion Blanks" and substitute "Fully Heat-treated Fin and Housing Stock"

Special Category: Delete "Special a, Hardness test, 80, 3.1, 4.5.3.1"

Note a: Delete in its entirety and substitute "(a) Four specimens total (2 round from housing stock, 2 flat from fin stock). Failure of any sample shall cause rejection of the applicable solution heat treatment batch or press-quench lot. See para. 4.4.1."

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Note b: Delete in its entirety and substitute "(b) Four specimens total (2 round, 2 flat). Failure of any sample shall cause rejection of the applicable solution heat treatment batch or press-quench lot. See para. 4.4.1."

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4.4.2.4, Housing

Major 116: Delete "4.5.5.2" and substitute "Visual" in Inspection Method column.

Minor 205: Delete "Gage" and substitute "Tool control/4.4.2.b" in Inspection Method column.

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* 4.4.2.6, Fin assembly: Add the following:

"Major 115, Blade security test, 5, 3.2, Gage/4.5.6."

PAGE 19

4.5.3.1, Hardness: Delete in its entirety.

4.5.3.1.1: Delete in its entirety.

4.5.3.2, Tension test: Delete in its entirety and substitute:

"4.5.3.2 Tension test. Following final heat treatment, 4 specimens (2 round from the cartridge housing stock, 2 flat from the fin stock) shall be taken from the pre-selected specimens representing the hottest and coldest zones of each solution heat treat batch or press-quench lot (see 4.4.1.1 and 4.4.1.2).

The flat tensile specimens shall be the largest standard flat-type specimens obtainable from the blade area, of fin stock and the remaining two specimens shall be the largest standard round-type specimens obtainable from housing stock.

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Specimens shall be taken longitudinally. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the applicable solution heat treatment batch or press-quench lot. The test criteria for all four tensile specimens shall be as prescribed by ASTM Method E-8. The number of specimens and their location is to verify compliance with the mechanical property requirements. The Government shall not be restricted from testing additional specimens at any location to determine that the mechanical properties meet the drawing requirements for yield strength and elongation."

4.5.3.2.1: Delete paragraph in its entirety.

4.5.3.2.2: Delete paragraph in its entirety.

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4.5.3.3, Microstructure examination: Delete paragraph in its entirety and add the following:

"4.5.3.3 Microstructure examination.

4.5.3.3.1 Fin stock. From the fin blade adjacent to each of the flat samples submitted for tension testing, one wafer approximately 1/2 inch square shall be prepared for metallographic examination from undeformed material. When suitably prepared, the specimens shall be etched with Keller's reagent and examined by scanning under a metallurgical microscope at a minimum magnification of 500 diameters. Overheating shall be evidenced by the presence of rosettes in the microstructure or incipient melting at the grain boundaries (see 6.5). Evidence of the above shall cause rejection of the applicable solution heat treat batch or press-quench lot.

4.5.3.3.2 Housing stock. Metallographic specimens from housing material shall be cut from a transverse section at either end of each round specimen submitted for tension testing. Preparation and examination for overheating, porosity etc. shall be as described in 4.5.3.3.1."

4.5.4 Hydrostatic test

line 10: Delete the following sentence: "The section where the fin is assembled to the housing shall be fully supported on the outside diameter during test."

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line 24: Delete the following sentence: "For first article inspection only, the test pressure shall be increased until the housing ruptures and the maximum pressure shall be recorded for information purposes."

PAGE 21

4.5.5.2: Delete paragraph in its entirety and substitute:

"4.5.5.2 Excessive flash holes. The sample fin housings shall be visually inspected for the presence of excess flash holes."

* 4.5.6 Blade security test add the following:

"4.5.6 Blade security test: At the start of production and whenever applicable tooling is changed or adjusted (inside diameter of fins, rear outside diameter or knurled portion of housing) 5 consecutive assemblies must pass the test. The housing portion shall be held securely in a fixture while 375 ± 25 inch-pounds torque is applied for 3 seconds to the blade portion of the assembly. Rotation of the blades around any housing shall be cause for rejection. If the blade cants are contacted during testing, the assembly must be discarded whether or not the assembly passes."

6.3, line 4: Delete: DRDAR-QAR-I and substitute DRDAR-QAT-I.

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* 6.8 Add the following: Direct submission to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAT-B, Dover, N.J. 07801.

The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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